# Tatiana Chakravorti

Postdoctoral Fellow, University of Alabama

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## Academic Appointments

**Postdoctoral Fellow**, University of Alabama 2026 - present
College of Communication and Information Sciences
**Assistant Professor**, KL University, India 2019 - 2020
Department of Electronics and Communication Engineering

## Education

**Ph.D., Informatics**, The Pennsylvania State University, University Park, PA, USA 2026
Advisor: Dr. Sarah Rajtmajer
**Ph.D., Electronics and Communication**, Siksha 'O' Anusandhan University, India 2019
Advisor: Prof. P. K. Dash
**M.Tech., Electronics and Communication Engineering**, WB University of Technology, India 2014
**B.Tech., Electronics and Communication Engineering**, WB University of Technology, India 2011

## Research Interests

Human-centered and responsible AI; human-AI collaboration and decision-making; AI ethics and governance; sociotechnical evaluation of AI systems

## Publications

### Peer-Reviewed Conference Proceedings

1. **Chakravorti, Tatiana**, Pranav Narayanan Venkit, Sourojit Ghosh, and Sarah Rajtmajer. "Beyond Detection: Governing GenAI in Academic Peer Review as a Sociotechnical Challenge." arXiv preprint arXiv:2603.20214 (2026)(Accepted to AIES 2026).
2. **Chakravorti, Tatiana**, Robert Fraleigh, Timothy Fritton, Christopher Griffin, Vaibhav Singh, Sai Koneru, C. Lee Giles, David Pennock, Anthony Kwasnica, and Sarah Rajtmajer. "Human-AI Collaboration for Estimating Scientific Replicability." Proceedings of the International Conference on Hybrid Human-Artificial Intelligence, vol. 423, pp. 161-175.
3. **Chakravorti, Tatiana**, Chuhao Wu, Sai Koneru, and Sarah Rajtmajer. "AI for research assessment: opportunities and challenges from India." Information Research an international electronic journal 31, no. iConf (2026): 1714-1732.
4. **Chakravorti, Tatiana**, Xinyu Wang, Pranav Narayanan Venkit, Sai Koneru, Kevin Munger, and Sarah Rajtmajer. "Social Scientists on the Role of AI in Research." In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, vol. 8, no. 1, pp. 528-540. 2025.
5. Gautam, Sanjana, Mohit Chandra, Ankolika De, **Tatiana Chakravorti**, Girik Malik, and Munmun De Choudhury. "Towards experience-centered AI: A framework for integrating lived experience in design and development." In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, vol. 8, no. 2, pp. 1062-1077. 2025.
6. Venkit, Pranav Narayanan, **Tatiana Chakravorti**, Vipul Gupta, Heidi Biggs, Mukund Srinath, Koustava

Goswami, Sarah Rajtmajer, and Shomir Wilson. "An Audit on the Perspectives and Challenges of Hallucinations in NLP." In Proceedings of the 2024 conference on empirical methods in natural language processing, pp. 6528-6548. 2024.

7. **Chakravorti, Tatiana**. "Enhancing transparency and research ethics through human AI techniques." In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, vol. 7, no. 2, pp. 8-10. 2024(Student paper).
8. Wu, Chuhao, **Tatiana Chakravorti**, John M. Carroll, and Sarah Rajtmajer. "Integrating measures of replicability into scholarly search: Challenges and opportunities." In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, pp. 1-18. 2024.
9. **Chakravorti, Tatiana**, Robert Fraleigh, Timothy Fritton, Michael McLaughlin, et al. "A prototype hybrid prediction market for estimating replicability of published work." In HHAI 2023: Augmenting Human Intellect: Proceedings of the Second International Conference on Hybrid Human-Artificial Intelligence, pp. 300-309. 1 Oliver's Yard, 55 City Road, London, EC1Y 1SP: SAGE Publications, 2023.
10. **Chakravorti, Tatiana**, Vaibhav Singh, Sarah Rajtmajer, Michael McLaughlin, Robert Fraleigh, Christopher Griffin, Anthony Kwasnica, David Pennock, and C. Lee Giles. "Artificial prediction markets present a novel opportunity for human-AI collaboration." arXiv preprint arXiv:2211.16590 (2022).
11. **Chakravorti, Tatiana**, Vinay Kumar Addala, and J. Shivam Verma. "Detection and classification of COVID 19 using convolutional neural network from chest X-ray images." In 2021 6th international conference for convergence in technology (I2CT), pp. 1-6. IEEE, 2021.
12. **Chakravorti, Tatiana**, and Penke Satyanarayana. "Classification of power quality disturbances using adaptive variational mode decomposition based random vector functional link network." In 2019 IEEE Region 10 Symposium (TENSYMP), pp. 721-726. IEEE, 2019.
13. Das, Debashreeta, **Tatiana Chakravorti**, and P. K. Dash. "Hilbert huang transform with fuzzy rules for feature selection and classification of power quality disturbances." In 2017 4th IEEE Uttar Pradesh Section International Conference on Electrical, Computer and Electronics (UPCON), pp. 439-445. IEEE, 2017.
14. **Chakravorti, Tatiana**, and P. K. Dash. "Morphology based fuzzy approach for detection & classification of simultanious power quality disturbances." In 2016 IEEE Annual India Conference (INDICON), pp. 1-6. IEEE, 2016.
15. **Chakravorti, Tatiana**, R. K. Patnaik, and P. K. Dash. "A morphological filter based disturbance detection and classification technique for DFIG wind farm based microgrid." In 2015 IEEE Power, Communication and Information Technology Conference (PCITC), pp. 979-985. IEEE, 2015.

**Peer-Reviewed Journal Articles**

1. Abatayo, A. L., Achakulvisut, T., Acuna, D., Aczel, B., Balaji, L., Bandrowski, A., Benjamin, D. M., Bishop, M. M., Brase, G. L., Brown, A. W., Bush, M., Caverlee, J., **Chakravorti, T.**, et al. (in press). Assessments of Credibility in the Social and Behavioral Sciences. Nature Communications.
2. **Chakravorti, Tatiana**, Sanjana Gautam, and Sarah Rajtmajer. "Open science practices among early-career human-computer interaction researchers in the US." Plos one 20, no. 11 (2025): e0334692.
3. **Chakravorti, Tatiana**, Sai Koneru, and Sarah Rajtmajer. "Reproducibility and replicability in research: What 452 professors think in Universities across the USA and India." PloS one 20, no. 3 (2025): e0319334.
4. **Chakravorti, Tatiana**, and Penke Satyanarayana. "Non linear system identification using kernel based exponentially extended random vector functional link network." Applied Soft Computing 89 (2020): 106117.
5. Bisoi, Ranjeeta, **Tatiana Chakravorti**, and Nihar Ranjan Nayak. "A hybrid Hilbert Huang transform and improved fuzzy decision tree classifier for assessment of power quality disturbances in a grid connected

distributed generation system." International Journal of Power and Energy Conversion 11, no. 1 (2020): 60-81.

6. **Chakravorti, Tatiana**, Lipsa Priyadarshini, P. K. Dash, and Badri Narayan Sahu. "Islanding and non-islanding disturbance detection in microgrid using optimized modes decomposition based robust random vector functional link network." Engineering Applications of Artificial Intelligence 85 (2019): 122-136.
7. **Chakravorti, Tatiana**, N. R. Nayak, Ranjeeta Bisoi, P. K. Dash, and Lokanath Tripathy. "A new robust kernel ridge regression classifier for islanding and power quality disturbances in a multi distributed generation based microgrid." Renewable Energy Focus 28 (2019): 78-99.
8. **Chakravorti, Tatiana**, and Pradipta Kishore Dash. "Multiclass power quality events classification using variational mode decomposition with fast reduced kernel extreme learning machine-based feature selection." IET Science, Measurement & Technology 12, no. 1 (2018): 106-117.
9. **Chakravorti, Tatiana**, Rajesh Kumar Patnaik, and Pradipta Kishore Dash. "Detection and classification of islanding and power quality disturbances in microgrid using hybrid signal processing and data mining techniques." IET Signal Processing 12, no. 1 (2018): 82-94.
10. **Chakravorti, Tatiana**, Rajesh Kumar Patnaik, and Praditpta Kishor Dash. "Advanced signal processing techniques for multiclass disturbance detection and classification in microgrids." IET Science, Measurement & Technology 11, no. 4 (2017): 504-515.
11. Nanda, Sarita, **Tatiana Chakravorti**, and P. K. Dash. "A new Taylor-LMS adaptive filter for parameter estimation of power signals including distributed generation systems." Australian Journal of Electrical and Electronics Engineering 13, no. 3 (2016): 174-194.
12. Nanda, Sarita, P. K. Dash, **Tatiana Chakravorti**, and Shazia Hasan. "A quadratic polynomial signal model and fuzzy adaptive filter for frequency and parameter estimation of nonstationary power signals." Measurement 87 (2016): 274-293.

**Reports, Preprints, and Manuscripts Under Review**

1. Sarah Rajtmajer, Laxmaan Balaji, D. M. Benjamin, James Caverlee, **Tatiana Chakravorti**, Yiling Chen, T. M. Errington, et al. “Innovations in Machine Assessment of Replicability.” OSF, 2026.
2. Alexandra Bejarano, Audrey DeHoog, Beyza Nur Guler, Yifei Zhu, **Tatiana Chakravorti**, and Qin Zhu; "Should Robots Be Compassionate? Undergraduate Perspectives on the Ethics of Compassionate Robots in Higher Education". (Under review)

## Grants And Funding

**Ethical Use of LLMs for Peer Review**, College of Information Sciences and Technology, Penn State $25,232

## Honors And Awards

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| 2025 | Graduate School Student Persistence Scholarship, Penn State ($1,100) |
| 2025 | CRA UR2PhD Graduate School Mentoring Scholarship ($1,000) |
| 2025 | AAAI/ACM AIES Student Program Award ($1,500) |
| 2024 | AAAI/ACM AIES Student Program Award ($1,200) |
| 2024 | Penn State Diversity Award to attend the Grace Hopper Celebration ($1,500) |
| 2024 | Travel Award, International Conference on the Science of Science and Innovation (ICSSI) ($1,500) |

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| 2023 | Honorable Mention, “Excellence at CAPWIC” Poster Award, ACM Capital Region Celebration of Women in Computing |
| 2022, 2023 | Travel Award, ACM Capital Region Celebration of Women in Computing (CAPWIC) ($500, twice) |
| 2022 | Travel Award, Metascience Conference ($1,500) |
| 2022 | IST Travel Award, AAMAS Conference ($1,800) |

## Invited Talks And Presentations

### Invited and Contributed Talks

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| 2025 | Invited Talk, Center for Socially Responsible Artificial Intelligence (CSRAI), Penn State. |
| 2024 | Lightning Talk, Association for Interdisciplinary Metaresearch and Open Science (AIMOS) Conference, Australia. |
| 2024 | Research Talk, International Research Culture Conference, UK (virtual). |
| 2024 | Research Talk, OpenFest Symposium “Towards an Open Research Culture,” University of Sheffield, UK (virtual). |
| 2024 | Research Talk, ACM Capital Region Celebration of Women in Computing (CAPWIC), University of Virginia. |
| 2023 | Lightning Talk, AIMOS Conference, Australia. |
| 2022 | "Designing Hybrid Crowd + AI Prediction Markets for Estimating Scientific Replicability." Human-Machine Collaboration Workshop, Paris, France. |

### Poster Presentations

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| 2024 | International Conference on the Science of Science and Innovation (ICSSI). |
| 2023 | "Hybrid Prediction Markets for Estimating Research Reproducibility." ACM CAPWIC, Virginia. |
| 2023 | "Human-AI Collaboration for Artificial Prediction Markets." Rao Conference, Penn State, State College, PA. |

## Research Experience

**Postdoctoral Fellow**, University of Alabama, USA 2026 - present

*Securing AI Research: A Lifecycle-Based Typology of Threats (NSF EAGER)*

- Developing a typology of security threats across the AI research lifecycle using Critical Incident Technique interviews with principal investigators, cybersecurity personnel, compliance and export control officers, research administrators, and federal partners, complemented by a national stakeholder survey.
- Combining qualitative analysis with NLP methods to identify threat patterns and build incident-grounded scenarios, to be validated with cross-sector stakeholders through a participatory workshop.

*Mapping AI Ethics Education (NSF Funded)*

- Contributing to a multi-institution, sequential explanatory mixed-methods study, including a national survey, mapping curriculum priorities, assessment strategies, and pedagogical approaches in AI ethics education across U.S. institutions.
- Examining how faculty perceptions and institutional needs and capacities shape AI ethics interventions, with partner networks spanning research universities, community colleges, and minority serving institutions.

*AI Ethics Bowl: Evaluating Students' Ethical Reasoning about AI (NSF Funded)*

- This NSF-funded project adapts the AI Ethics Bowl into a classroom approach for teaching AI ethics in undergraduate STEM courses, using real-world case studies and team debate. A pre/post evaluation showed students gained familiarity with AI and ethics and moved from naming AI harms to reasoning through ethical principles such as autonomy, justice, and responsibility.
- Developing a semi-structured interview protocol to examine how team deliberation and public argumentation shape students' ethical judgment and views on delegating decisions to AI.

**Graduate Research Assistant**, The Pennsylvania State University, USA 2021 - 2026

**Responsible AI for Peer Review** 2025 - 2026

- Conducted an interview study with AI journal editors and reviewers, followed by quantitative analysis, examining how they use, evaluate, or avoid large language models (LLMs) in peer review and the implications for judgment, effort, and research ethics.

**AI for Synthesizing Social Science Knowledge** 2024 - 2025

- Used interviews and surveys to map the opportunities and ethical concerns social science researchers associate with AI in their disciplines.

**DARPA Systematizing Confidence in Open Research and Evidence (SCORE)** 2021 - 2022

- Developed hybrid prediction markets in which human participants trade alongside AI-enabled agents to forecast the replicability of published research.
- Designed human–AI collaboration features for an AI-assisted replicability prediction tool based on researcher needs, and conducted usability testing of the tool.
- Conducted interviews and observations with researchers on the reproducibility crisis and open science practices.
- Designed and statistically analyzed a survey of geographically diverse researchers.

**Research Assistant**, Siksha 'O' Anusandhan University, India 2015 - 2018

*Microgrid Disturbance Detection*

- Developed signal processing and machine learning methods for detecting and classifying disturbances in microgrids.

**Project Assistant Intern**, University of Applied Sciences Augsburg, Germany 2017

*LINDA Project (Local Island Power Supply with Distributed Generation Systems in Case of Large-Scale Blackouts)*

- Analyzed field data from local island power supply systems during large-scale blackout scenarios.

## Teaching Experience

**Graduate Teaching Assistant**, College of Information Sciences and Technology, Penn State 2022 - 2026

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| Spring 2026 | Concepts and Emerging Issues in Data Privacy (Foundational Course) |
| 2023 - 2024 | Language, Logic, and Discrete Mathematics (Fall 2023 - Fall 2024) |
| Fall 2022 | Privacy and Security for Data Sciences |
| Spring 2022 | Foundations of Data Privacy |
| Spring 2022 | Seminar for Data Science |

**Assistant Professor**, KL University, India 2019 - 2020

Taught courses in machine learning, communication systems, and introductory Python programming.

Supervised undergraduate (B.Tech.) capstone projects in teams of two, including:

- Real-world anomalous activity detection in surveillance videos using deep learning.
- Power quality pattern recognition and classification using the Hilbert-Huang transform and fuzzy logic.
- Detection and classification of COVID-19 from chest X-rays using convolutional neural networks (published at IEEE I2CT 2021).

## Professional Service

### Peer Review

Reviewer for ACM CHI, ACM CSCW, ACM FAccT, AAAI/ACM AIES, and AIMOS (2022 - present).

### Organizing and Leadership

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| 2024 | Organizing Committee Member, Workshop on Research Integrity, Penn State |
| 2024 | Organizing Committee Member, AIMOS Conference, Australia |
| 2024 | Student Affiliate, Center for Socially Responsible Artificial Intelligence, Penn State |
| 2019 - 2020 | Departmental Coordinator, International Wing, KL University, India |
| 2019 - 2020 | Course Coordinator and Library In-charge, Department of Electronics and Computer Science, KL University, India |
| 2015 | Organizing Committee Member, IEEE Power, Communication and Information Technology Conference (PCITC), India |

## Methods And Technical Skills

**Qualitative methods:** interviews, observation, focus groups, participatory design, usability testing, thematic analysis, content analysis

**Quantitative methods:** survey design, statistical analysis, machine learning, artificial intelligence

**Programming and tools:** Python, R, SQL, SPSS, MATLAB/Simulink, C.